Aditi Dankar

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Profile Summary

Machine Learning professional with 2+ years of software engineering and 1.5+ years of research experience in building Deep Learning models. Specializing in Computer Vision and Generative AI. Proven ability in designing and implementing novel multimodal AI models with expertise in Diffusion Models and Transformer architectures.

Education

MSc Machine Learning for Visual Data Analytics (Vision & Language) *Sep 2024 – Sep 2025*
Queen Mary University of London, London, UK

- Grade: Distinction
- *Dissertation:* Trajectory-Conditioned Multimodal Diffusion Model — Designed a novel framework for controllable 3D motion (dance) synthesis conditioning a diffusion model on simultaneous audio and spatial trajectory inputs. Introduced a custom trajectory loss function and a new evaluation metric, reducing trajectory error 71.8% from baseline. *View Project*
- *Key Modules:* Deep Learning & Computer Vision, Neural Networks & NLP, Machine Learning for Visual Data Analytics, Introduction to Computer Vision, Natural Language Processing, Machine Learning.

BTech Electrical and Electronics Engineering *Aug 2014 – Jul 2018*
Manipal Institute of Technology, Manipal, India

Indian Institute of Science (IISc), Continuing Education *2020 – 2023*
Completed graduate-level coursework in Machine Learning, Deep Learning, and Mathematics for ML.

- *Key Modules:* Deep Generative Models (A+), Optimization Foundations for Machine Learning (A), Probability, Statistics, and Matrix Methods for Machine Learning (A), Mathematics for Machine Learning (B+).

Professional Experience

Research Assistant, Indian Institute of Science, Bengaluru, India *Jan 2023 – Dec 2023*

- Architected and benchmarked end-to-end computer vision pipelines for non-destructive PCB analysis using YOLOv8 and Segment Anything Model (SAM).
- Achieved 94% precision and 84% recall in component detection using PyTorch, improving inspection accuracy.

ML Research Intern, Tata Institute of Fundamental Research, Mumbai, India *Jan 2022 – Jun 2022*

- Designed and implemented a custom CNN-LSTM architecture using TensorFlow to classify pulsating stars from structured spectral density data, enabling improved signal interpretation in astrophysics datasets.
- Achieved 83% recall, demonstrating effective application of deep learning to astrophysics signal classification.

Project Engineer – Frontend Developer, Wipro Limited, Bengaluru, India *Jul 2018 – Sep 2020*

- Developed and deployed production-level web applications and collaborated within Agile/Scrum teams, contributing to sprint planning, stand-ups, and CI/CD pipelines.

Technical Skills

- **NLP & LLMs:** LLM Evaluation & Fine-Tuning, Transformer Architectures (BERT, T5), Attention Mechanisms, Text Generation, Summarisation, Dialogue Systems, RNNs
- **Computer Vision & Multimodal:** Generative Models (Diffusion, GANs), Object Detection (YOLO), Segmentation, Multi-modal Data Processing (Audio, Video, Text), CNNs
- **Frameworks & Libraries:** PyTorch, Hugging Face, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy
- **Programming & Tools:** Python, PDB, HPC cluster, WandB, SQL, Git, Anaconda, LaTeX, Linux/Unix

Projects

Aspect-Based Sentiment Analysis with BERT

- Fine-tuned pre-trained BERT models for the task of aspect-based sentiment analysis.
- Developed and evaluated an extended architecture using LSTM layers on top of BERT embeddings to capture more nuanced contextual information, achieving 82% classification accuracy on MAMS-ATSA dataset.

Abstractive Summarisation & Text Generation using T5

- Fine-tuned T5 on the XSum dataset for abstractive summarisation and for keyword-to-text generation, evaluating summarisation quality using ROUGE metrics.

Conversational Dialogue System with Attention

- Developed a GRU-based seq2seq model with attention mechanism on the Cornell Movie-Dialogs Corpus, analyzing attention behavior and decoding strategies (greedy vs. beam search).

Transformer Implementation for Image Classification

- Implemented core multihead self-attention and cross-attention mechanisms from scratch in PyTorch, integrating them into a Transformer architecture for image classification.

Diffusion Models for Image Generation

- Implemented and trained forward/reverse sampling pipelines, exploring denoising strategies and training dynamics.

Publications

- A. Dankar, P. P. Kundapur, "Automated Mobile Attendance System (AMAS)", IEEE ICAC3, 2019. *Link*